SET - A

Name of the Course : B.Sc.(H) Computer Science

Name of the paper : Design and Analysis of Algorithms

Vishu

Semester : IV

Date- 17th March 2026 **Duration: 1 Hour** **Maximum Marks: 30**

Attempt all questions

Q1 a) Which sorting algorithm out of insertion and quick-sort will sort an array of size n faster if the list is already sorted? Why? 2

b) Discuss the running time of the following snippet of code: 3

```
count = 0
for(i=1,i<=n),i++)
        for(j=1,i<=n),j = 2*j)
                count++
```

O(nlogn)

10

Q2 Sort the array {7, 3, 2, 28, 11} using the **Selection sort, and Merge Sort** techniques and write the status of the array during every iteration of each pass. Also, find the cumulative number of comparisons at each iteration. 5

Q3 Use Strassen's algorithm to compute the product of the following matrices : 5

$$\begin{pmatrix}2 & 3\\5 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\2 & 2\end{pmatrix}$$

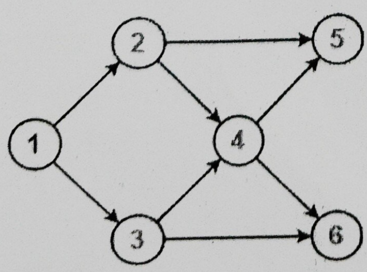
$\begin{bmatrix}8 & 8\\7 & 7\end{bmatrix}$

Q4 Given the set of following activities, find the set of activities and thus the number of activities that can be performed by a person, given that he/ she can perform only one task at a time: 5

Activity Name	Start time	Finish time
A	1	2
B	3	4
C	0	6
D	5	7
E	8	9
F	5	9

What will be the criteria of picking up the activities? Which algorithmic approach you followed?

Q5 Demonstrate the procedure for finding any one topological order for the following graph, clearly showing the graph at each step. Also, find all possible topological orderings. 5



1 2 3 4 5 6

Q6 State and describe the two properties of Greedy approach of problem solving 5

Set- B

Vishu

Name of the Course : B.Sc.(H) Computer Science

Name of the paper : Design and Analysis of Algorithms

Semester : IV

Date- 17th March 2026 **Duration: 1 Hour** **Maximum Marks: 30**

Attempt all questions

Q1 a) For each of the following sorting algorithms, merge sort and insertion sort, discuss whether or not it is (i) stable. (ii) in-place 3

b) State the cut property of graph. 2

Q2 Demonstrate the working of Insertion **sort, Quick-Sort** algorithms on the array { 3, 7, 9, 12, 15}. write the status of the array during every iteration of each pass. Also, find the cumulative number of comparisons at each iteration. 5

Q3 Use Strassen's algorithm to compute the product of the following matrices: $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 4 & 2 \\ 6 & 1 \end{pmatrix}$ 5

Q4 Illustrate the Counting sort algorithm on the following array: 5
A= {2, 4, 1, 2, 9, 1, 5, 8, 2, 3, 8, 9, 5, 2, 1}

Q5 Show the result of running BFS on the following graph (Figure-1) given below using the vertex D as source. Show the tree after each step 5

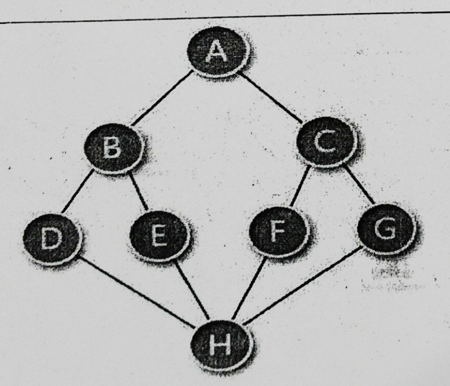


Figure 1: Graph for BFS Traversal

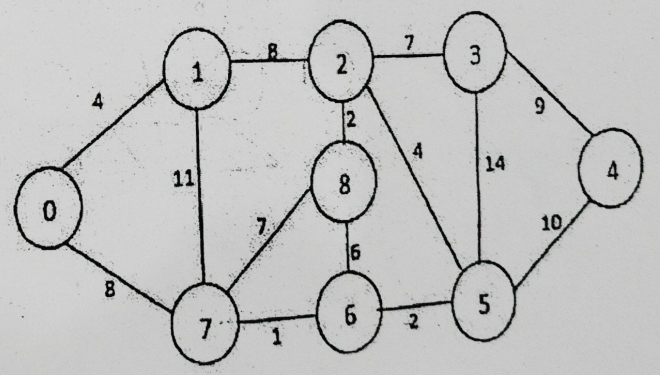


Figure 2: Graph for MST-Prim's Algo

Q6 For the graph given in Figure 2, find a minimum spanning tree using the Prim's algorithm considering node 1 as the start node. Also, state the time complexity of the same. 5